

S2 2026 FIT5137 Assignment 2 Specification (Weight = 40%)

Due date: Thursday, 17 September 2026 11:55pm

Version: 4.0 – 18/08/2026

A. General Information and Submission

- ☐ This is an individual assignment.
- ☐ This assignment requires the Oracle database. You can connect using the Oracle service provided by Monash ([Week 1 setup](#))
- ☐ *Submission method*: Submission is online through Moodle.
- ☐ *Penalty for late submission*: 5% deduction for each day.
- ☐ *Assignment FAQ*: There is an [Assignment Frequently Asked Questions page](#) set up for the Assignment 2 on EdStem Forum.
- ☐ A marking rubric has been provided for Assignment 2: [A2 Marking guide](#)

B. Problem Description

MonCity is one of the largest smart cities in the world, and was developed by Monash University for education and research purposes. In MonCity, there are four different zone areas (ZoneA, ZoneB, ZoneC, ZoneD). A wide range of self-driving cars is provided for Monash students and staff members to travel between different zone areas or within the same zone area.

MonCity has an [existing operational database](#) that *maintains and stores all of the self-driving car-related information*, such as the booking, maintenance, and accident records required for management's daily operation. However, in order to improve work efficiency, management at MonCity has decided to hire your team of data warehouse engineers to design and develop a data warehouse that can quickly generate reports based on their needs. Management at MonCity wants to generate reports to keep track of the **bookings, accidents and maintenance information**, such as calculating statistics of booking records and accident records, which can be used for self-driving car analyses later. For the **accident** details, they are particularly interested in the **number of accidents per self-driving car** (e.g., the number of accidents caused by Car01).

MonCity's operational database tables can be found in the **MonCity** account. You can, for example, execute the following query:

Select * From MonCity.<table_name>;

The data definition of each table in MonCity is as follows:

Table Name	Attributes and Data Types		Notes
FACULTY	FACULTYID	VARCHAR	This table stores faculty information. One faculty can have multiple passengers.
	FACULTYNAME	VARCHAR	
	ZONE	VARCHAR	
RESEARCHCENTER	CENTERID	VARCHAR	This table stores information about research centres. One research centre may have several maintenance teams.
	CENTERNAME	VARCHAR	
	PHONE	NUMBER	
	OPENINGHOUR	VARCHAR	
PASSENGER	PASSENGERID	VARCHAR	This table stores passenger information. Each passenger is from a certain faculty and can have one or multiple booking records.
	PASSENGERNAME	VARCHAR	
	PASSENGERROLE	VARCHAR	
	PASSENGERGENDER	VARCHAR	
	PASSENGERAGE	NUMBER	
	FACULTYID	VARCHAR	
ERROR	ERRORCODE	VARCHAR	This table stores accident error information. One error code may involve more than one accident.
	ERRORMESSAGE	VARCHAR	
MAINTENANCETYPE	MAINTENANCETYPE	VARCHAR	This table stores maintenance type information. One maintenance type may involve more than one maintenance record.
	MAINTENANCEDESCRIPTION	VARCHAR	

BOOKING	BOOKINGID	VARCHAR	This table stores the information of booking records. One booking record belongs to a certain passenger and car.
	REGISTRATIONNO	VARCHAR	
	BOOKINGDATE	DATE	
	DEPARTUREZONE	VARCHAR	
	DESTINATIONZONE	VARCHAR	
	PASSENGERID	VARCHAR	
CAR	REGISTRATIONNO	VARCHAR	This table stores the information of self-driving cars. One car may have more than one booking, one accident, and one maintenance record.
	CARMODEL	VARCHAR	
	MANUFACTURINGYEAR	NUMBER	
	CARBODYTYPE	VARCHAR	
	NUMSEATS	NUMBER	
MAINTENANCE	MAINTENANCEID	VARCHAR	This table stores each car's maintenance history. Each maintenance record belongs to a certain car, maintenance type and maintenance team.
	REGISTRATIONNO	VARCHAR	
	MAINTENANCEDATE	DATE	
	MAINTENANCETYPE	VARCHAR	
	MAINTENANCECOST	NUMBER	
	TEAMID	VARCHAR	
ACCIDENTINFO	ACCIDENTID	VARCHAR	This table stores the information of each accident, including accident ID, accident zone, etc. Each accident can be categorised into a certain error code.
	ACCIDENTZONE	VARCHAR	
	CAR_DAMAGE_SEVERITY	VARCHAR	
	ERRORCODE	VARCHAR	
CARACCIDENT	REGISTRATIONNO	VARCHAR	This table stores information about accidents and all related cars. One
	ACCIDENTID	VARCHAR	

			accident can involve more than one car.
MAINTENANCETEAM	TEAMID	VARCHAR	This table stores information about maintenance teams. One maintenance team may have more than one maintenance record. Each maintenance team may be from more than one research center.
	TEAMLEADER	VARCHAR	
	PHONE	NUMBER	
BELONGTO	TEAMID	VARCHAR	This table stores the relationship between research centres and maintenance teams.
	CENTERID	VARCHAR	

C. Task

The assignment is divided into **FIVE** main tasks:

C1.Design a data warehouse for the above MonCity database.

You are required to create a data warehouse for the **MonCity** database. The management is interested in the following fact measures:

- Number of booking records
- Total maintenance cost
- Number of maintenance records
- Number of accident records

The following are some possible dimension attributes that you may need in your data warehouse:

- Booking time period: by month
- Faculty
- Passenger age group: Young adults (18-35 years old); middle-aged adults (36-59 years old); old-aged adults (60 years old and over)
- Car body type (including number of seats)
- Research centre
- Maintenance type

- Accident Zone (based on accident zone in the ACCIDENTINFO table)
- Error code
- Car damage severity

For each attribute, you may apply your own design decisions on specifying a range or group where this is not already detailed above, but make sure to specify them in your submission.

C2. Preparation stage.

You must ensure that you have thoroughly explored the operational database before proceeding with the data warehouse design. This includes **understanding the tables, attributes, and relationships, drawing an ERD** to represent the operational database structure, and **performing sufficient data cleaning** to ensure the data is suitable for the data warehouse. Once you have done the data cleaning process, you are required to **explain what strategies you have taken to explore and clean the data**

The outputs of this task for **Report** are:

- If you have done the data cleaning process, explain the strategies you used in this process (you need to show the SQL to explore the operational database and SQL of the data cleaning, as well as the screenshot of data *before* and *after* data cleaning).

C3. Drawing a star/snowflake schema.

The star schema for this data warehouse may contain multi-fact(s). You need to identify the fact measures, dimensions, and attributes of the star/snowflake schema. The following queries might help you to identify the fact measures and dimensions:

- How many booking records were made by passengers from the IT faculty in July by using bus?
- How many booking records were made by each passenger age group?
- How many maintenance records were listed that include at least one maintenance team from research centre 4 (CE04)?
- List the total maintenance cost of each maintenance type for mini bus.
- How many accidents were recorded in the ZoneA for Car01?
- List the number of accidents for each error code at ZoneB for Car06.
- List the number of accidents with severe damage inflicted on Car05.

Note:

The star or snowflake schema you design **may or may not require bridge tables or determinant dimensions**. If bridge tables are necessary, make sure to include the attributes **GroupList** and **WeightFactor** in the relevant dimension table. If you are using determinant dimensions, make sure the correct notation has been used (broken lines for the table boundary) and you must provide the reasons for your choice(s).

You should pay **attention to the granularity** of your fact tables. You are **required to create two versions** of star/snowflake schemas based on **different levels of aggregation**. **Version-1** should be in the **highest level of aggregation**. **Version-2** should be in **level 0, which means it contains no aggregation**.

Version Name	Level
Version-1	Highest aggregation
Version-2	No aggregation (Level 0)

The outputs of task A & B for **Report** are:

- Two versions of star/snowflake schema diagrams. (You can use Lucidchart or Draw.io to draw the star schema.)

Note:

The star/snowflake schema design, including the fact measures and dimensions, is entirely based on the specific questions we provide for each exercise and assignment. It is essential to closely follow the question requirements and design the star schema accordingly. The fact measures and dimensions should align precisely with the data and analysis needed to address the given problem.

Please ensure that you thoroughly understand the question prompts and the expected outcomes before proceeding with the star schema design. **Any deviation from the provided question may lead to inaccurate results and impact your assessment.**

Asking the tutor to check whether your design is correct or not will not be answered, as it is part of the assessment in this assignment.

C4.Implement the **Version-1** of the star/snowflake schemas using SQL.

You are required to implement the Version-1 of star/snowflake schema that you have drawn in Task 1. This implies that you need to create the fact and dimension tables in SQL. The output is a series of SQL statements to perform this task. You will also need to show that this task has been carried out successfully.

Note:

- If your account is full, you will need to drop all of the tables that you have previously created during labs.
- If you have dropped all tables in your account and you still encounter the ORA-01536: space quota exceeded for tablespace 'TABLE_NAME', please check your SQL code whether you have properly joined all tables. This issue was mainly caused when you did not do the table join properly as the number of records multiplied during the process.

The outputs of this task for **Report** are:

- Screenshots of the table structure you created for Version-1, including the dimension tables and fact tables.

A sample of screenshots of the table structure

Columns	Column Name	#	Type	Type Mod	Not Null	Default	Comment
Columns	ADDRESS_ID	1	NUMBER		[]		
Constraints	STREET_NUMB	2	NUMBER		[]		
Foreign Keys	STREET_NAME	3	VARCHAR2(100)		[]		
References	SUBURB	4	VARCHAR2(20)		[]		
Triggers	STATE	5	VARCHAR2(10)		[]		
Indexes	POSTCODE	6	NUMBER		[]		
Privileges							
Dependencies							
Statistics / ...							
DDL							
Virtual							

C5. Data Analysis and Visualisations

Due to the successful operation of the MonCity project, Monash University decided to invest further in the self-driving car project. As a data warehouse engineer:

- You are required to provide a **suggestion with valid supporting data** to help management **decide on how to improve current self-driving car projects**, such as *which car body type should be further invested in based on the Booking, maintenance, and accident records, or which error code has contributed to the most number of accidents, etc.*; or
- If you have a **different opinion on the self-driving car project**, **provide a reason why the project is not worth further investment.**

Note:

1. The above suggestion must contain **valid supporting data from the MonCity data warehouse**.
2. Create appropriate **visualisations (e.g., charts or graphs)** in Power BI [Web version] to support the argument and recommendation presented in C5. Your

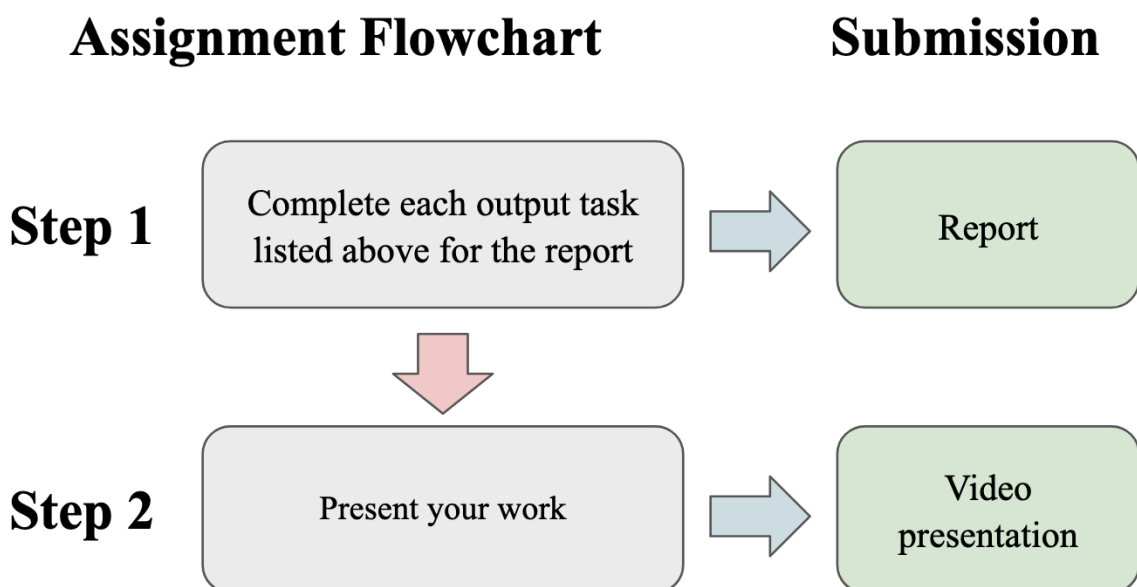
visualisations should clearly communicate the key findings from your data analysis. This could be tables, graphs, or charts.

3. Please use a maximum of 500 words for your suggestions.

The outputs of this task for **Report** are:

- **Findings report: A detailed explanation of your findings, including any significant observations or patterns identified during the analysis with sufficient visualisation.**
- You can take screenshots of your Power BI visualisations and place them alongside the corresponding explanations in your report.

D.Submission Checklist



Step 1: Report (40% of the total score)

A combined pdf file save as: YourstudentID_A2_report.pdf, containing all of the above tasks:

- A. Cover page
- B. If you have done the data cleaning process, explain the strategies you used in this process (you need to show the SQL to explore the operational database and SQL of the data cleaning, as well as the screenshot of data *before* and *after* data cleaning).
- C. Two versions of star/snowflake schema diagrams. (You can use Lucidchart or Draw.io to draw the star schema.)
- D. Screenshots of the table structure you created for Version-1 only, including the dimension table and fact tables.
 - a. SQL file for creating the star/snowflake schema is **NOT** required in submission

- E. Findings report: A detailed explanation of your findings, including any significant observations or patterns identified during the analysis with sufficient visualisations.

Step 2: Video presentation (60% of the total score)

A five minute video presentation in mp4 format save as: YourstudentID_A2_video.mp4

Based on the report you have created, present your design and findings in a **five-minute video presentation**.

E. Assignment Submission

The assignment must be submitted electronically through Moodle. Please ensure the following:

1. Step 1 output: A combined pdf file save as: YourstudentID_A2_report.pdf
2. Step 2 output: A five minute video presentation in mp4 format save as: YourstudentID_A2_video.mp4

Zip all above files from step 1 to 2, and name the ZIP folder as A2_YourstudentID.zip.

The due date: **Due - Thursday, 17th September 2026, 11:55 PM**

- The submission of this assignment must be in the form of a **single ZIP file**. **Only PDF and .mp4 files will be accepted within the zip file. No other formats will be accepted.**
- You must ensure that you have all the files listed in this checklist before submitting your assignment to Moodle. Failure to submit a complete list of files will lead to mark penalties.
- It is important to note that our support hours are limited and we don't have the capacity to deal with submission issues outside of working hours. You must ensure that you have all the files listed in this checklist before submitting your assignment to Moodle. Failure to submit a complete list of files will result in a mark penalty.
- Penalty for late submission: 5 % deduction for each day, including weekends.
- Submission Cut-off time: **24th September 2026, 11:55 PM** (Submission link will be unavailable after the cut-off date).

F. Authorship

This assignment is an **individual assignment** and the final submission must be identifiably your own work. Breaches of this requirement will result in an assignment not being accepted for assessment and may result in disciplinary action.

G.Late Penalty:

Late assignments submitted without an approved extension may be accepted up to a maximum of **seven days** with the approval of the Chief Examiner and/or Lecturer but will be **penalised at the rate of 5% per day (including weekends and public holidays)**. Assignments submitted more than seven days after the due date will receive a zero mark for that assignment and may **not receive any feedback**.

Please note:

- An inability to manage your time or computing resources will not be accepted as a valid excuse. (Several assignments being due at the same time are a fact of university life.)
- Hardware failures, whether of personal or university equipment, are not normally recognised as valid excuses. Failure to back up assignment files is also not recognised as a valid excuse.

H.Special Consideration:

All extensions / special considerations will now be handled by the central Spec Con team. **Please do not email teaching staff to request an extension or special consideration.**

Extensions and other individual alterations to the assessment regime will only be considered using the University Special Consideration Policy. Students should carefully read the [Special Consideration website](#), especially the details about what formal documentation is required.

All special consideration requests should be made using the [Special Consideration Application](#).

Please do not assume that submission of a Special Consideration application guarantees that it will be granted – you must receive an official confirmation that it has been granted.

I.Getting help and support:

What can you get help for?

- ***Consultations with the Teaching Team***

Talk to the Teaching Team:

<https://learning.monash.edu/course/view.php?id=31420§ion=5>

- ***English language skills***

Talk to English Connect: <https://www.monash.edu/english-connect>

- ***Study skills***

Talk to a learning skills advisor: <https://www.monash.edu/library/skills/contacts>

- ***Counselling***

Talk to a counsellor: <https://www.monash.edu/health/counselling/appointments>

G.Plagiarism and Collusion:

Monash University is committed to upholding standards and academic integrity and honesty. Please take the time to view these links.

[Academic Integrity Module](#)

[Student Academic Integrity Policy](#)

[Test your knowledge, collusion \(FIT No Collusion Module\)](#)

All the best for your Assignment!